

Computer Vision Platformer

Plan of approach

Team Members: Razvan Apostu,

Solly Nankin,

Andrew Cardoso,

Ivan Fuentes

Contents

Introduction	3
Project definition.....	3
Levels layout	6

Introduction

The purpose of this project is to demonstrate the students' ability to integrate a full product using professional organizational methods such as class diagrams, state diagrams and system diagrams, all using the UML language. The project will use computer vision elements paired with pygame to create a unique game idea.

Project definition

The project is a 2D platformer game that is linked with an external camera with which the user can import objects into the game to aid in their progression through the levels.

This game aims to provide unique user experience, and a new way to play video games. This game will interact with physical geometric shapes (squares, triangles, circles) detected by a webcam which will import those objects in the game to aid the player into jumping on high platforms, blocking attacks or solving puzzles.

User story:

As a player, I want to use computer vision, so that I can import objects into the game to progress.

- User Requirements

The user requirements will be categorized using the “MoSCoW” method.

- ➔ Must: A requirement that is mandatory for the project
- ➔ Should: A requirement that is not mandatory, but is ideal
- ➔ Could: A requirement that improves the quality of the product, consider if extra time is available
- ➔ Wont: A requirement that must not be fulfilled, recommended to omit this one, and instead rely on “Must” requirements

1. General requirements: Requirements that are general on all projects, the baseline.

General Requirements			
UR	Description	MoSCoW rating	Approved
1	The project will be primarily coded using Python	Should	Yes
2	The platformer will be developed using pygame	Must	Yes
3	The camera detection will be developed using OpenCV	Should	Yes
4	Communication between devices needs to be done using MQTT	Must	Yes
5	The game has any keyboard/mouse input	Must	Yes
6	Minimum of 10 OOP classes used within project (can be inherited from pygame/OpenCV libraries)	Must	Yes
7	Game provided with state diagram as documentation	Must	Yes
8	Game has title screen	Could	Unknown
9	Game has pause feature	Could	Unknown

2. World requirements: Requirements concerning the aesthetics of the levels, and how they're laid out

World Requirements			
UR	Description	MoSCoW rating	Approved
1	World has gravity	Must	
2	Platforms have custom sprites	Should	
3	Game has multiple levels	Must	
4	Game has final boss fight level	Must	
5	Fade-in/Fade-out animation when transitioning between levels	Should	
6	Game has "endurance mode" which will be the	Could	

	boss fight getting progressively harder		
7	World has a “dungeon” aesthetic	Could	
8	Boss fight, has a “block” event when the player needs to place an object to block an attack	Should	
9	Don’t request the same shape twice in a row between blast stages	Should	
10	Boss fight needs to be challenging and fast paced	Must	

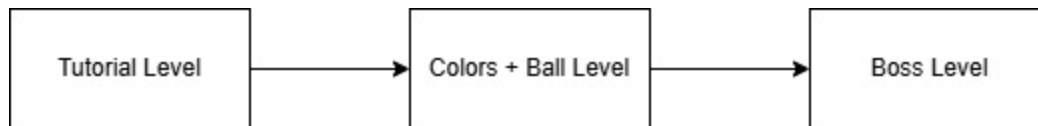
3. Player requirements: Requirements that have to do with the in-game player model such as how it behaves relative to human input.

In-game player requirements			
UR	Description	MoSCoW rating	Approved
1	Player will be controlled using arrow keys	Should	
2	Player movement is responsive and fast	Should	
4	Player has no HP system. If it gets hit by an object, an instant respawn is triggered	Should	
5	Player can drop down from a high platform to a lower platform by pressing the “down” key	Should	
6	Pressing the down key while mid air will trigger a “ground slam” where the player quickly goes down	Could	
7	Player has custom sprite	Should	
8	Player has movement animation	Should	

4. Camera requirements: Requirements regarding object detection

Camera Requirements			
UR	Description	MoSCoW rating	Approved
1	Detection speed as fast as possible	Must	
2	Camera frame should be small enough to only allow one object in the frame at a time	Should	
3	Camera needs to detect both shape type and color	Must	
4	Camera needs to detect a square, a triangle and a circle	Must	

Levels layout



- Tutorial Level

The level consists of 2 platforms: Ground + an upper platform

The player needs to get up to the second platform by placing a cube in the camera frame.

The cube will spawn on the map and act as a step stool for the player to jump to the second platform.

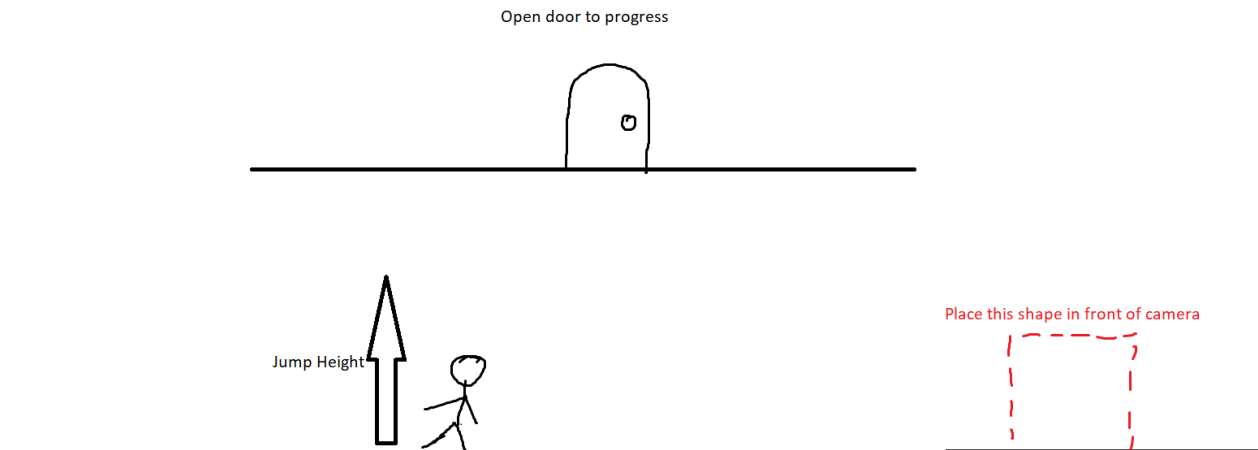
The second platform has a door that transitions to the next level.

***The tutorial level may have text indicating the controls and mechanics**

Rough sketch of the tutorial level

Arrow keys to move
Pressing down arrow while mid-jump
will trigger a SLAM

LEVEL 1/3

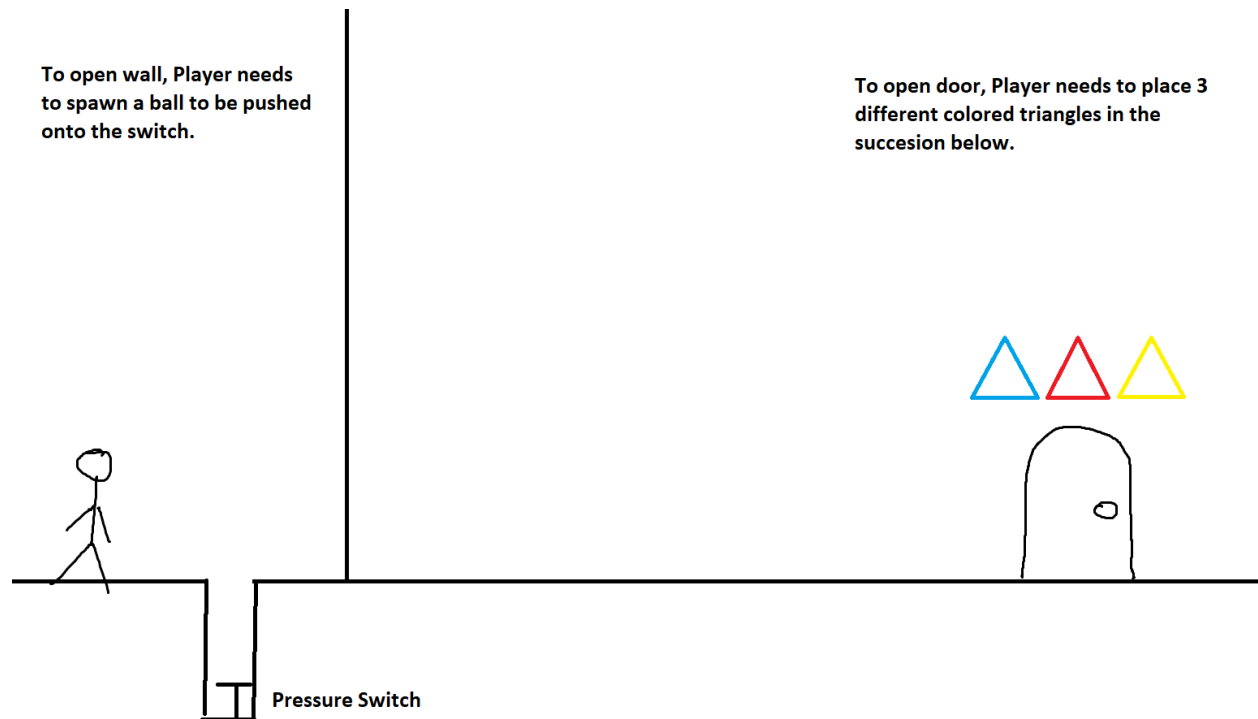


- **Puzzle Level**

This level is a simple puzzle to demonstrate the capabilities of image detection.

This level consists of a single platform (Ground), an unlockable wall and an unlockable door to progress to the next level.

In this level, the player won't receive any tips indicating the ball needs to be spawned and pushed into the hole to press the switch.



- Boss level

The boss fight consists of 3 platforms: Ground, Platform 1 and Platform 2.

The player can jump and descend between platforms without needing extra blocks as in the tutorial

The boss fight will last for a specific time (ex. 60 seconds)

During this fight there will be 2 stages:

1. Spike stage

Spikes will randomly generate.

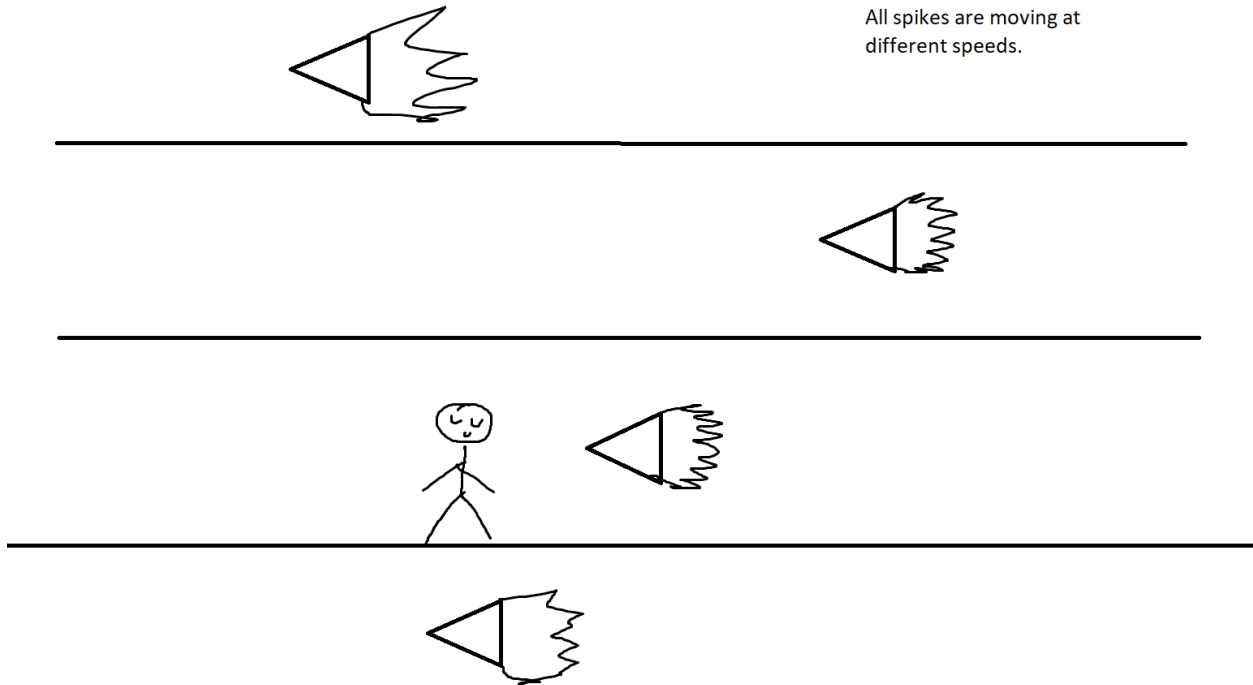
They will move from right to left on either platform, their speed will also be randomized.

If the spike hits the player, player dies

The spikes can only be dodged in this stage

TIME LEFT: 37s

All spikes are moving at
different speeds.



2. Huge blast stage

This stage is a quick time event meant to challenge the player's reaction.

Before this stage gets triggered the player will receive an **exclamation mark** on his screen indicating that the blast is coming.

Simultaneously an outline of a square, circle or triangle will randomly spawn on one of the platforms.

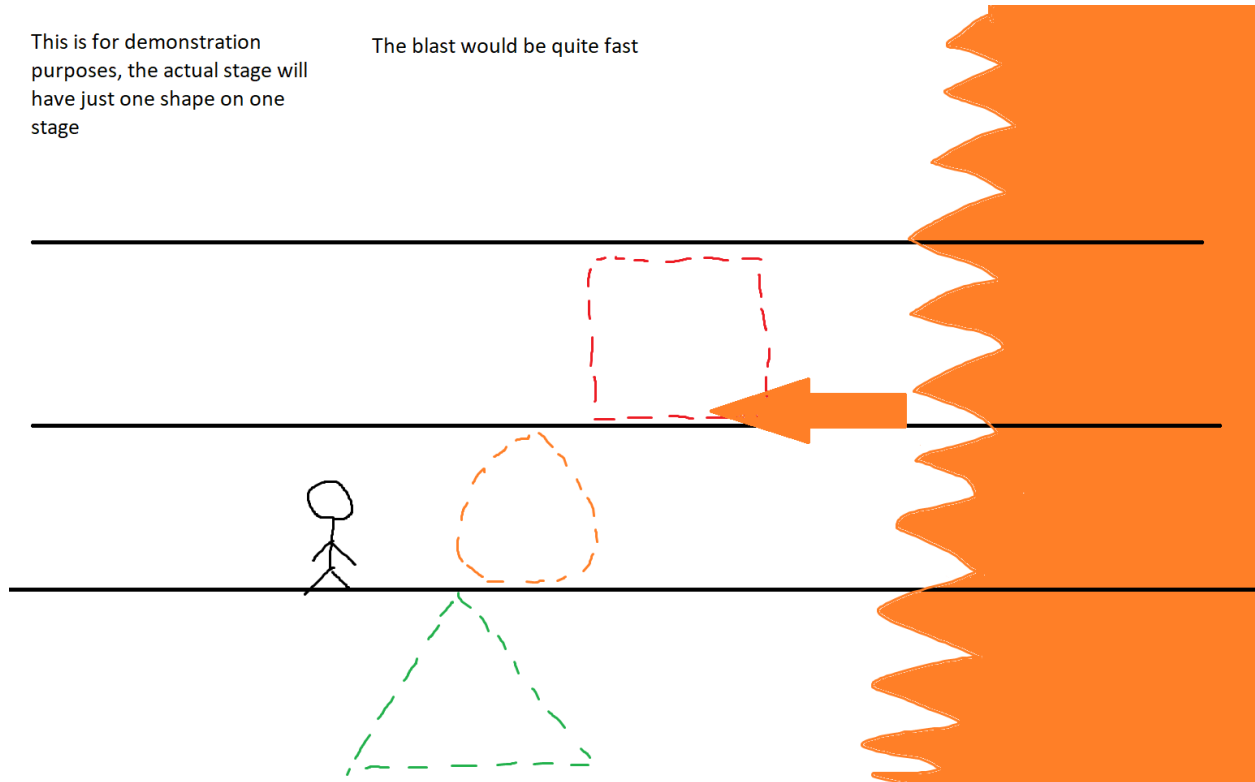
The only way for the player to survive the blast is to place the requested object in front of the camera and to stand behind it after it's spawned in the game.

Software Design & Architecture

MC3B1

This is for demonstration purposes, the actual stage will have just one shape on one stage

The blast would be quite fast



*****THIS BOSS FIGHT IS MEANT TO BE CHALLENGING AND FAST PACED EVEN FOR EXPERIENCED GAMERS. So, the spikes should spawn at a rapid rate at a relatively high speed. Also, the big blast is supposed to stress out the player by giving little time to process the situation and react accordingly.**